



ME 101 INTRODUCTION TO MECHANICAL ENGINEERING

Ali Fethi Okyar (Sec1)
Can Polat Çıgay (Secs 2-4)

FALL 2026

- Welcome to the Fall 2026 semester! All students are expected to enroll to the course at the YULEARN (moodle) website. Enrollment key for the course TBA.
- Students must obtain a @std.yeditepe.edu.tr extension email address for signing up to YULEARN. To get your email address send an email containing your student number, name, surname to labadmin@cse.yeditepe.edu.tr.

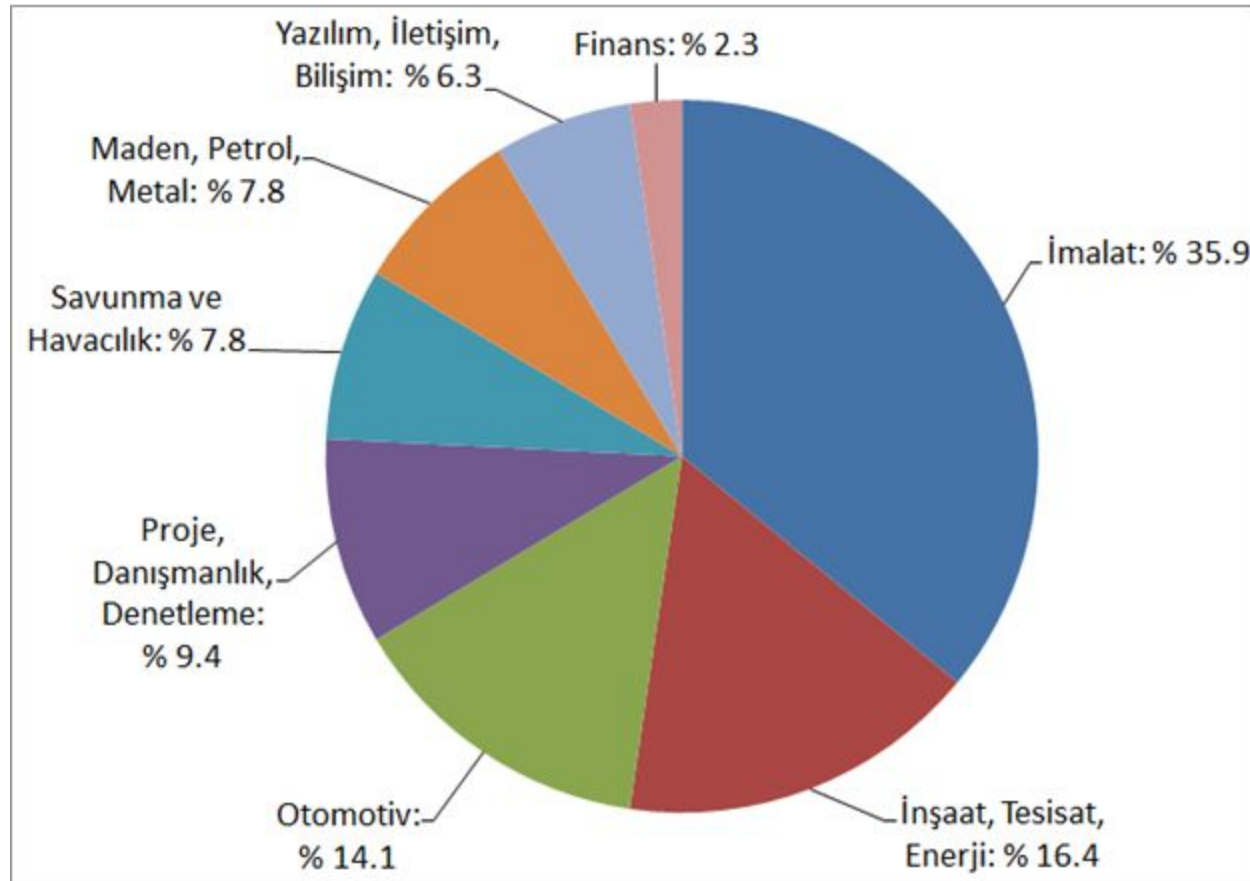
OUR DEPARTMENT

- Mechanical Engineering Department was founded in 2001
- 11 instructors (öğretim elemanı; does not necessarily imply rank), 6 RAs
- Currently ~500 undergraduate students
- >1000 graduates* till now
 - First graduates in 2005
- 6 laboratories
- One workshop (atölye)

Ranks in academia:

- Instructor (öğretim görevlisi)
- Assistant professor (Dr. Ogr. Uyesi)
- Associate professor (doçent)
- Professor (“full professor”) (profesör)

OUR GRADUATES (Based on a survey of 200 grads)



OUR FACULTY

- Founded in 1996 (when 7tepe was first founded)
- Faculties are responsible for **undergraduate education (lisans eğitimi)**
- The name of our faculty:
 - Until 26.8.2014:
→ MÜHENDİSLİK ve MİMARLIK FAKÜLTESİ
 - After 26.8.2014:
→ MÜHENDİSLİK FAKÜLTESİ
 - After 2025:
MÜHENDİSLİK ve DOĞA BİLİMLERİ FAKÜLTESİ
- ~3500 undergraduate students in 10 departments
(excluding the Preparatory School)

OUR FACULTY

Yeditepe Üniversitesi Mühendislik ve Doğa
Bilimleri Fakültesi



LINKEDIN
QR
CODE

FACULTIES AT THE UNIVERSITY

- There are 13 faculties at Yeditepe
 - Some of the faculties:
 - Architecture
 - Fine Arts (Güzel Sanatlar),
 - Law (Hukuk),
 - Medicine (Tıp),
 - Communication (İletişim),
 - etc.

INSTITUTES (ENSTİTÜLER)

- Institutes are responsible for **graduate education (lisansüstü eğitim)**: masters level(yüksek lisans) and doctoral (PhD) level
- There is one institute at Yeditepe
 - Institute of Graduate Education (Yüksek Lisans Enstitüsü)

ADMINISTRATORS

- Head of a department: chairman, chairperson, chair (bölüm başkanı)
- Head of a faculty: dean (dekan)
 - **Faculty Executive Board (Fakülte Yönetim Kurulu):**
authority to decide about daily issues about students
→ Course transfers (muafiyet / ders sayma), semesterly leaves of absence (kayıt dondurma), graduations (mezuniyet), etc.
- Head of an institute: institute director (enstitü müdürü)
- Head of a university: rector (Europe),
president (USA), rektör (Turkish)
- Vakıf üniversitelerinde en üst düzeydeki yetkili birim / kişi:
Mütevelli Heyeti / Mütevelli Heyet Bşk.

WEIGHT OF A COURSE

Needed: a measure of the weight of a course

- One **credit-hour** (or, one **credit**) :

one **contact hour** (50 min) per week.

→ **contact hour:** 50 min of lecture **or**
100 min of lab/practice work

- ECTS : **E**uropean **C**redit **T**ransfer
and Accumulation **S**ystem

(each credit corresponds to about 25 hours of work)

(AKTS: **A**vrupa **K**redi **T**ransfer **S**istemi)

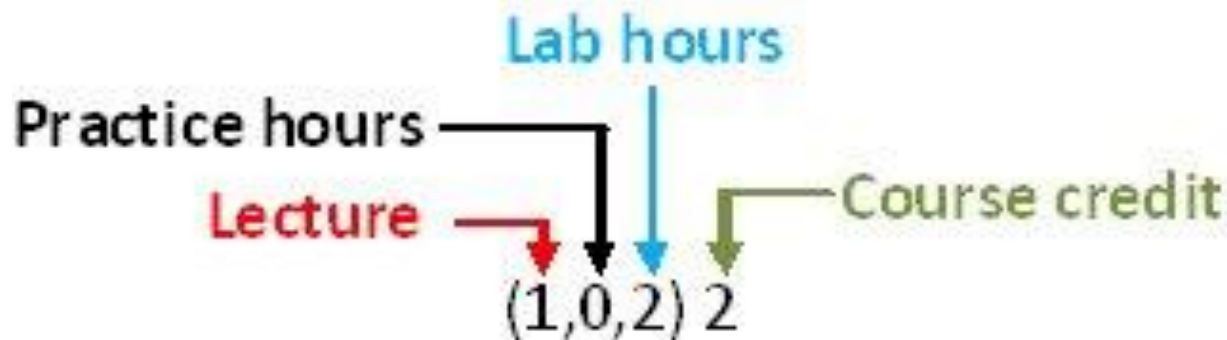
→ More complicated:

contact hours+time spent outside classroom, ...

Credit Structure of ME 101

- $(1,0,2) 2 \rightarrow \rightarrow$ A two credit-hour course

Shortly : A two credit course



- ECTS of ME 101 : 5
- The **total** number of **ECTS** credits is **240** for all four-year undergraduate programs.

CONDUCT OF THE COURSE

There will be

- one hour of lecture and
- two hours of laboratory (“lab”) work

every week.

- There will be a joint section for lectures and four lab sections.

CONDUCT OF THE COURSE

- Rule: at least 80 % attendance is required.
⇒ **Don't miss more than 20 %.**
- You **cannot miss more than two lab sessions.**
- Miss more than 20 % total or more than 2 labs
⇒ ⇒ ⇒
 - you get FA,
 - you cannot take the final exam,
 - you have no right to the makeup exam (bütünleme)⇒ ⇒ ⇒ **(You fail the course)**

CONDUCT OF COURSES IN ME DEPARTMENT

- **Do not sign** the attendance sheet **on behalf of an absent friend**
- Do not be late for the class
⇒ if the classroom door is closed, **knock to enter**
- **Do not enter if it is more than 5 minutes after the hour**

IN-CLASS BEHAVIOR IN ME DEPARTMENT

- Listen, Draw, Write, Engage
- Make it a habit to **take notes during the lecture**

⇒ **Bring to class**

- a notebook,
- pen, pencil, eraser
- a calculator
- a mini ruler

⇒ Carry these in a small **backpack**

IN-CLASS BEHAVIOR IN ME DEPARTMENT

- **No cell phone use.** Make sure your phone is
 - turned off, or
 - in silent mode.

(if what is going on in class is not interesting to you, you don't need to be there)

- No exchange of pencils, erasers, calculators, etc. **in exams. Bring your own!**
- **Do not** attempt to **cheat** in exams.

WRITING RULES IN ME DEPARTMENT

- Use size A4 paper not torn from a notebook.
- **Mark the margins** of the paper on all four sides and **do not write in the margins** (a couple of centimeters on all sides)
- Do not use pens. **Use pencils!**
- The idea behind homework is for you to learn the subject matter. If you **copy homework** from your friends, **the only person you hurt is yourself.**
- If your homework is a copy, you will get zero and disciplinary action may be taken against you
- Turn in your homework in due time. Late ones will either not be accepted or they will be marked down.

- “Engineer” : ingeniosus (Latince), ingenious (İngilizce)
 - a skillful and smart person who invents or builds a clever tool/equipment/machine (becerikli, zeki, hünerli, akıllıca icat ya da imalat yapan)
- “Engine”: a clever and useful equipment (akıllıca yapılmış yararlı aygıt)

ENGINEERING

© Original Artist
Reproduction rights obtainable from
www.CartoonStock.com



"Art, art, art - When are we going to get some engineers?"

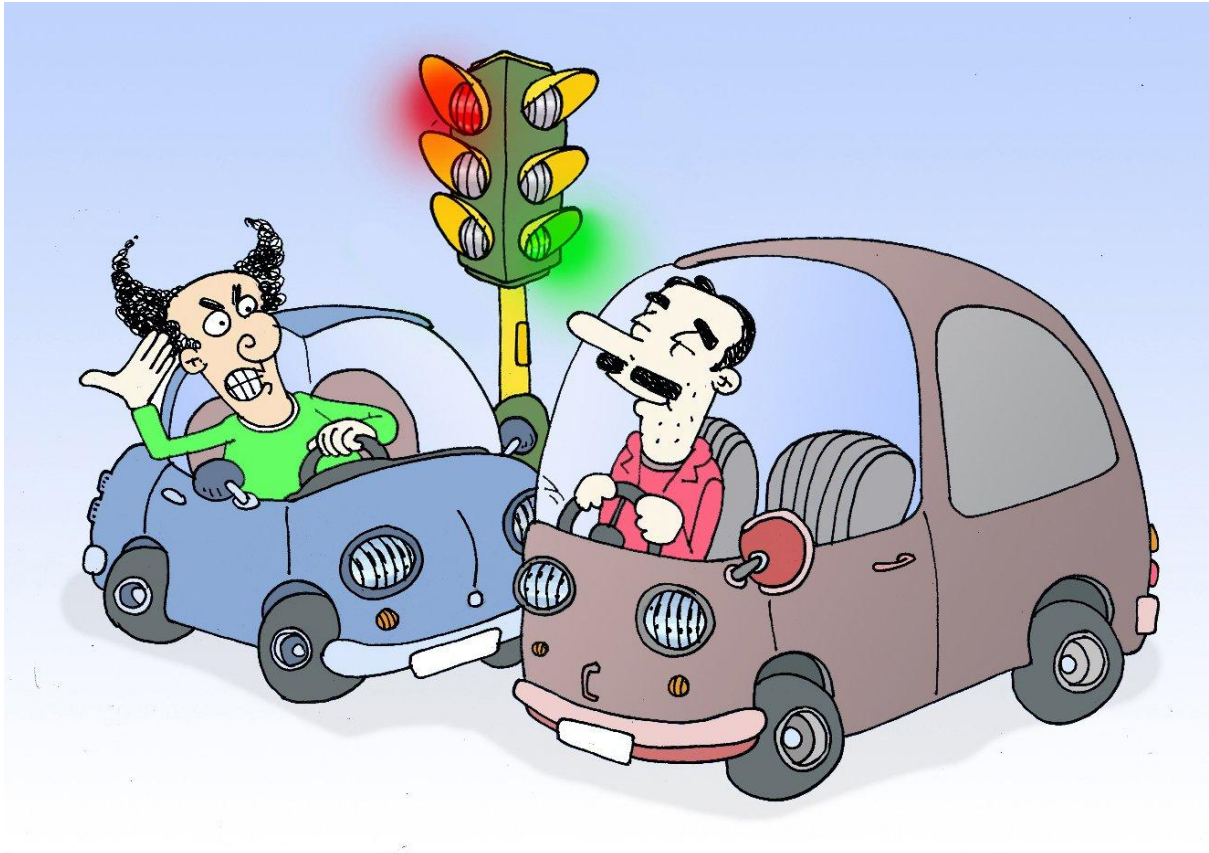
ENGINEERING =MÜHENDİSLİK

- Hendese = geometri
- From the word hendese → mühendis

- Engineers apply the theories and principles of **science and mathematics** to find economical solutions to technical problems.
- Mühendislik: **fen bilimleri ve matematiğin**, bir gereksinimi karşılayacak şekilde uygulanarak bir sistem, süreç, aygıt tasarlanıp imal edilmesi.

- Men had to be “engineers:
→ hunting, defense, tool making, ...
- “Engineering” began in ancient times.
→ clever inventions:
wedge, lever, wheel, pulley, ...

- The diplomas that we grant must have substance (Verdiğimiz diplomanın altı dolu olmalı)



- Each graduate is an ambassador of our department; a representative of our future graduates.
- It is **hard to gain** a reputation of “a department with high quality graduates”, but **very easy to loose** it.
- We have to maintain a certain level of quality.
- You **must deserve to graduate**.

YOU MUST WORK HARD!

SUCCESS/FAILURE AS A STUDENT

A few reasons for failure
in your first year in the department:

- **TYT-AET** fatigue $\Rightarrow$ laziness
- Not knowing the chosen field (ME) closely.
- “I already know math and physics”.
- To have a weak foundation (temeli zayıf olmak).
-

SUCCESS/FAILURE AS A STUDENT

- A common scenario of repeated failure (“Ben zaten matematik, fizik biliyorum”.)
 - İki arkadaşın sohbeti (yer: Mühendislik kantini)
 - A- Sınıfta hoca türev, limit filan anlattı.
 - B- Dersaneden bildiğim konular; çalışmama gerek yok, biraz baksam yeter.

SUCCESS/FAILURE AS A STUDENT

→ → Just because some math or physics terms are familiar to you, **do not think that you know the subject!!!**

- Unfortunately, today high school and “dersane” education teaches you **how to solve a problem** in 1-2 minutes.
- As a result, in most cases you do not understand **why you solve the problem that way**.
- Engineering requires you to **know the WHY!**

SUCCESS/FAILURE AS A STUDENT

→ 16 hafta sonra

A- Calculus'ten ne aldın?

B- F!!!

A) Hani biliyordun dersi.

B) Valla ben de anlamadım

SUCCESS/FAILURE AS A STUDENT

- University physics and math education:
 - Aims to teach you the “**why of phenomena**”.
 - Aims to **train your brain** so that you learn to think like an engineer.
 - **Analytical thinking, ability to reason.**
 - So that you can analyze problems and find solutions in your future career.
- **In your future career as an engineer:**
 - No one will give you 5 possible solutions and ask you to choose one.
 - You will be asked to **figure out** (find) **a solution on your own**, or even, to **find a non-existent solution**.

SUCCESS/FAILURE AS A STUDENT

→ Bir dönem sonra (yer: çimler) :

A- Calculus nasıl gidiyor?

B- Bir sürü soru çözdüm, AA hedefliyorum.

→ 14 hafta sonra:

A) Calculus ne geldi?

B) F!!!

SUCCESS/FAILURE AS A STUDENT

- Gaining the ability to think like an engineer:
 - Not possible by **ONLY** trying to solve (**OR**, to “read” the solutions of) as many problems as you can!
 - **OR**, by listening to a friend solving problems
- You **must understand** the underlying principles (**the why**).
 - ⇒ **must study the subject matter** in your text book (**kitaptan konuyu çalışmalısınız!!!**)
 - Then attempt to solve problems...

SUCCESS/FAILURE AS A STUDENT

→ İki dönem sonra (yer: Kadıköy'de bir kafe)

A- Calculus çalışıyor musun?

B- Sabah akşam soru çözüyorum.

→ 14 hafta sonra:

A) Calculus notun?

B) F!!

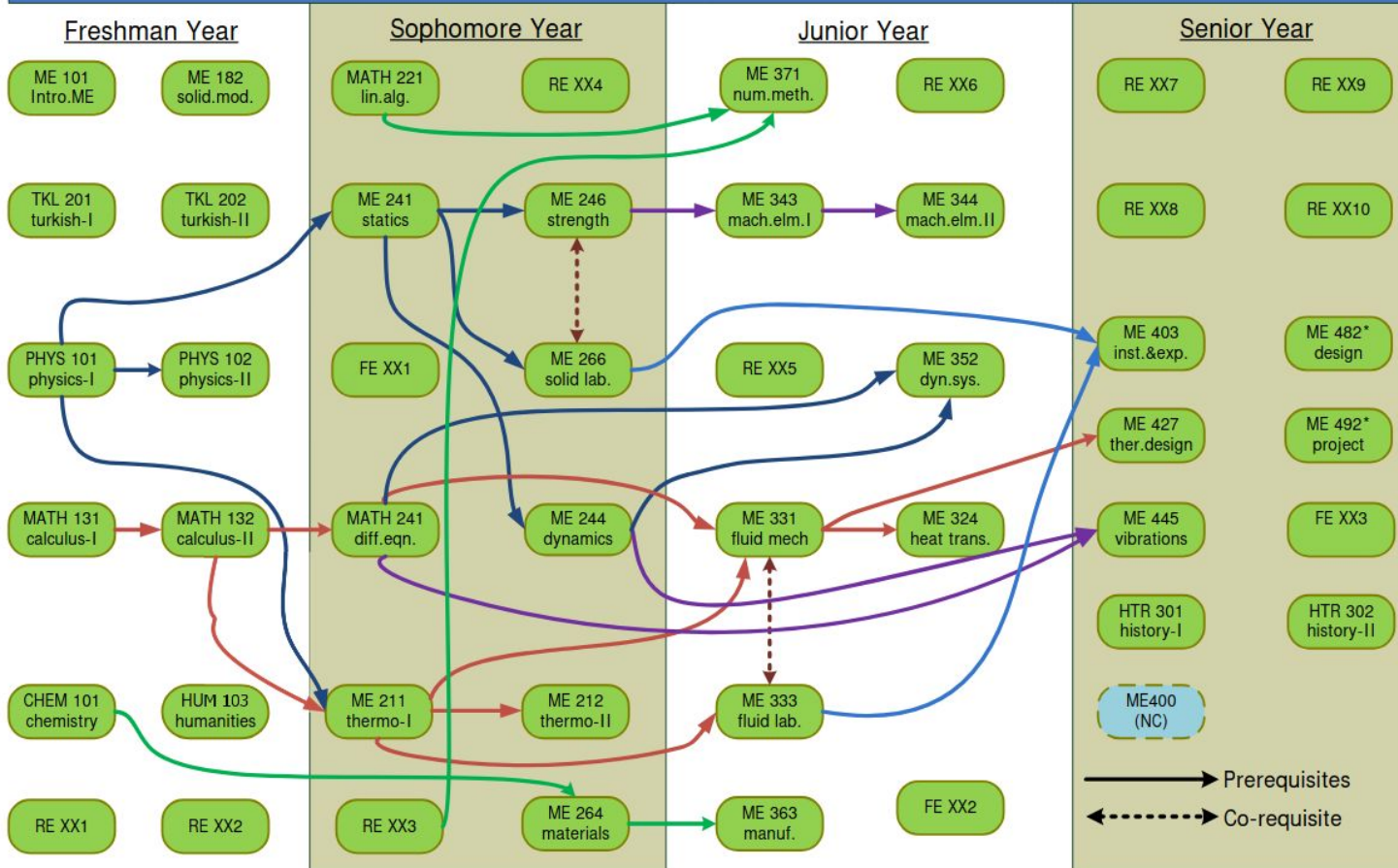
SUCCESS/FAILURE AS A STUDENT

Some reasons that **will get you in trouble later:**

- **Prerequisite chain (önkoşul zinciri):**
→ (You cannot build a structure without a **foundation**)
- Have barely passed MATH and PHYSICS courses (“hasbelkader” geçmiş olmak)
→ **lack of adequate knowledge**
- Got DD and DC from high-credit courses
→ **CGPA problem (ortalama sıkıntısı)**

PREREQUISITE CHAIN

MECHANICAL ENGINEERING DEPARTMENT CURRICULUM AND PREREQUISITES



* Sadece mezuniyet durumundaki öğrenciler ME 482 ve ME 492 derslerini alabilir

* Only students who are in their final semester for potential graduation can take ME 482 and ME 492.

01.09.2024

ME 101, Fall 2024 (güncel için bölüm web sayfası)

EXAMPLES OF PREREQUISITE CHAINS

- **MATH 131 → MATH 132 → MATH 241 → ME 331
→ ME 352
→ ME 445**
- **PHYS 101 → ME 241 → ME 246 → ME 343
→ ME 344**
- **MATH 131 → MATH 132 → ME 211 → ME 331
→ ME 324**

CALCULATION OF GPA (Grade Point Average)

Each letter grade has a corresponding **multiplier**:
AA→4.00, BA→3.50, BB→3.00, ..., DD→1.00, FF→0

- You get DC from a 4-credit course: **quality points** you earn is $1.50 \times 4 = 6.00$
- Add all quality points you have earned in a semester
- Divide the total quality points (36) by the total number of credits (20) you took that semester
⇒ you find your **semesterly GPA** (say, 1.80),
- Do this for all the semesters till now to find **CGPA**
(Cumulative GPA = Cumulative Grade Point Average)

(Genel Not Ortalaması = **GNO** in Turkish)

SUCCESS/FAILURE AS A STUDENT

The price you pay for passing high-credit courses with low grades:

Example:

Chem 101 (**4**) : DD

MATH 151 (**4**) : DD

PHYS 101 (**4**) : DD

ME 101 (2) : BB

AFE 131 (3) : BB

TKL 201 (3) : BB

$$\Rightarrow \text{GPA} = 36/20 = 1.80$$

(If it were a simple average:

3 DDs and 3 BBs $\Rightarrow$ the average would be CC (2.00)

It is a **weighted average!**

Weighted by the course credits!

SUCCESS/FAILURE AS A STUDENT

Example: A student in his/her **3rd** year
(**NOT** “3. year”!) who has taken
70 credits worth of courses so far.

- Say, s/he has a cumulative GPA of 1.75
- Means she collected a total quality points of
CGPA*completed credit = $1.75 * 70 = 122.5$

SUCCESS/FAILURE AS A STUDENT

Example (continued).

- Currently, s/he takes a course load of 15 credits:
In order to bring his/her **CGPA to 2.00** at the end of the current semester, s/he must accumulate a total of

$$(70+15)*2.00=\mathbf{170} \text{ quality points}$$

→ S/he already has **122.5**

⇒ During the current semester s/he must earn **$170-122.5=47.5$** quality points

⇒ His/her **GPA for the current semester** must be **$47.5/15=3.17$** . (> BB)

MORAL OF THE STORY

Low grades in the first two years
are harder to compensate for
in later years!!!

SUCCESS/FAILURE AS A STUDENT

- Aim a high CGPA; do not just barely pass your courses.
- Toward that aim, study hard beginning in your first year. **4-credit courses** are in the first two years.
- A high CGPA is useful for
 - Getting the most prestigious jobs
 - Getting good scholarships for graduate study
 - For getting achievement scholarship (**başarı bursu**) from Yeditepe.

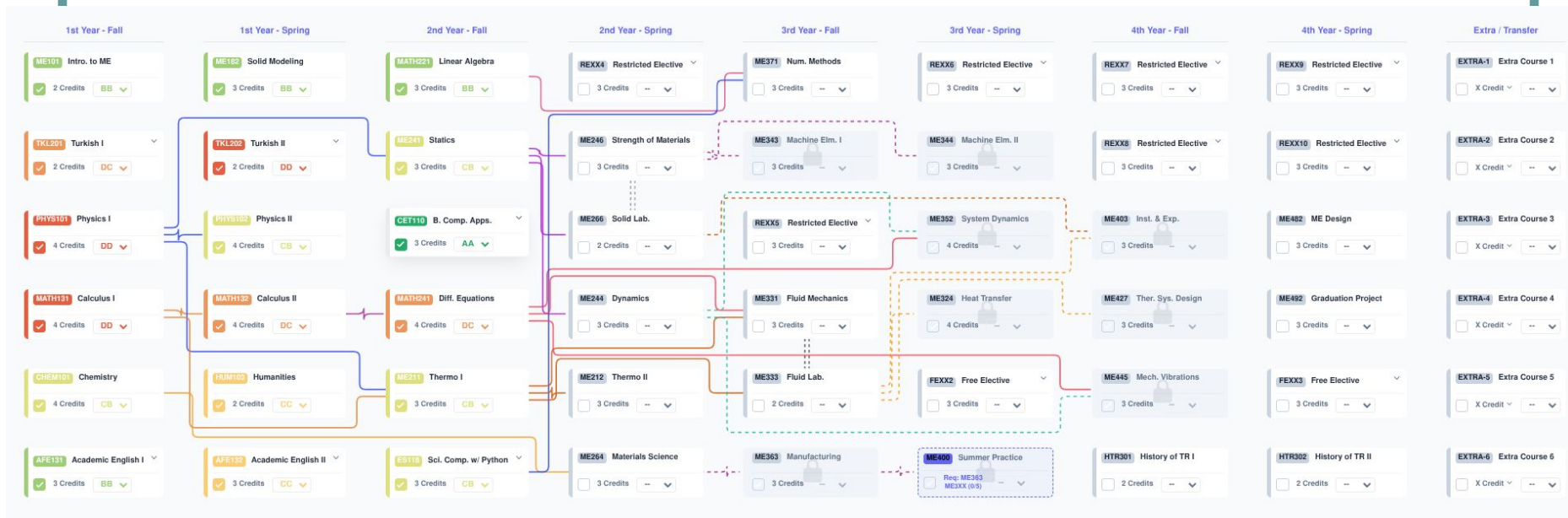
And, it might make you feel good, proud of yourself

SOME USEFUL ADVICE

- Know the curriculum (“müfredat”) well.
- Know the prerequisites well.
- Learn how to calculate your CGPA.
- Keep an **up-to-date excel file for the courses** you have taken.

The ME Curriculum Planner

<https://fen1kks.github.io/Interaktif-Akademik-Planlayici/>
(designed and implemented by Hızır Ketenci, ME'28)



SOME USEFUL ADVICE

- Kayıt
- Ders ekleme bırakma (“add-drop”)
- Dersten çekilme (“withdrawal”)
- Ders saydırma, dilekçe işleri
- Son tarihlere dikkat etmek
- Panoları izlemek

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Kayıta önce F ve W dersleri
- Çakışma en çok 1 saat: daha fazlasında ders bırakılmak zorunda — PANOLAR

- E-dönüşümdeki (OBS) bilgileriniz doğru olsun: e-posta, cep tel ****

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Bölümünüzün web sayfasını izleyiniz
DUYURULAR****
- Yönetmelikler:
ÜNİVERSİTE ANA SAYFASI****
- Yönergeler
 - Sınav Yönergesi
 - Yapay Zeka Politika Belgesi (hazırlanıyor)

ÖNEMLİ HUSUSLAR ve ÖNERİLER

- Sunuş becerileri
- Lisan becerileri

TÜRKÇE ve İNGİLİZCE*****

- Serbest seçmeliler